

Introduction to Graph Theory: Basic Concepts and Graph Types

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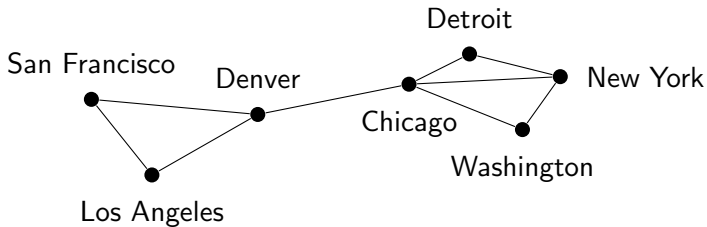
Definition

A **graph** $G = (V, E)$ consists of V , a nonempty set of **vertices** (or **nodes**) and E , a set of **edges**. Each edge has either one or two vertices associated with it, called its **endpoints**. An edge is said to **connect** its endpoints.

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The set of vertices V of a graph G may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an **infinite graph**, and in comparison, a graph with a finite vertex set and a finite edge set is called a **finite graph**.



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So far the graphs we have introduced are **undirected graphs**.
Their edges are also said to be **undirected**.

Definition

A **directed graph** (or **digraph**) (V, E) consists of a nonempty set of vertices V and a set of directed edges (or arcs) E . Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to **start** at u and **end** at v .

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- A graph with both directed and undirected edges is called a **mixed graph**.

TABLE 1 Graph Terminology

Type	Edges	Multiple Edges Allowed?	Loops Allowed?
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Graph Models

Acquaintanceship and Friendship Graphs. We can use a simple graph to represent whether two people know each other, that is, whether they are acquainted, or whether they are friends (either in the real world or in the virtual world via a social networking site such as Facebook). Each person in a particular group of people is represented by a vertex. An undirected edge is used to connect two people when these people know each other, when we are concerned only with acquaintanceship, or whether they are friends.

Graph Models

Influence Graphs In studies of group behavior it is observed that certain people can influence the thinking of others. A directed graph called an influence graph can be used to model this behavior. Each person of the group is represented by a vertex. There is a directed edge from vertex a to vertex b when the person represented by vertex a can influence the person represented by vertex b . This graph does not contain loops and it does not contain multiple directed edges.

Collaboration Graphs. A collaboration graph is used to model social networks where two people are related by working together in a particular way. Collaboration graphs are simple graphs, as edges in these graphs are undirected and there are no multiple edges or loops. Vertices in these graphs represent people; two people are connected by an undirected edge when the people have collaborated. There are no loops nor multiple edges in these graphs. The Hollywood Links graph is a collaborator graph that represents actors by vertices and connects two actors with an edge if they have worked together on a movie or television show.

Call Graphs. Graphs can be used to model telephone calls made in a network, such as a long-distance telephone network. In particular, a directed multigraph can be used to model calls where each telephone number is represented by a vertex and each telephone call is represented by a directed edge. The edge representing a call starts at the telephone number from which the call was made and ends at the telephone number to which the call was made. We need directed edges because the direction in which the call is made matters. We need multiple directed edges because we want to represent each call made from a particular telephone number to a second number.

The Web Graph. The web can be modeled as a directed graph where each web page is represented by a vertex and where an edge starts at the web page a and ends at the web page b if there is a link on a pointing to b . Because new web pages are created and others removed somewhere on the web almost every second, the web graph changes on an almost continual basis.

Citation Graphs. Graphs can be used to represent citations in different types of documents, including academic papers, patents, and legal opinions. In such graphs, each document is represented by a vertex, and there is an edge from one document to a second document if the first document cites the second in its citation list. (In an academic paper, the citation list is the bibliography, or list of references; in a patent it is the list of previous patents that are cited; and in a legal opinion it is the list of previous opinions cited.) A citation graph is a directed graph without loops or multiple edges.

Airline Routes. We can model airline networks by representing each airport by a vertex. In particular, we can model all the flights by a particular airline each day using a directed edge to represent each flight, going from the vertex representing the departure airport to the vertex representing the destination airport. The resulting graph will generally be a directed multigraph, as there may be multiple flights from one airport to some other airport during the same day.

Road Networks. Graphs can be used to model road networks. In such models, vertices represent intersections and edges represent roads. When all roads are two-way and there is at most one road connecting two intersections, we can use a simple undirected graph to model the road network. However, we will often want to model road networks when some roads are one-way and when there may be more than one road between two intersections. To build such models, we use undirected edges to represent two-way roads and we use directed edges to represent oneway roads. Multiple undirected edges represent multiple two-way roads connecting the same two intersections. Multiple directed edges represent multiple one-way roads that start at one intersection and end at a second intersection. Loops represent loop roads. Mixed graphs are needed to model road networks that include both one-way and two-way roads.

Round-Robin Tournaments. A tournament where each team plays every other team exactly once and no ties are allowed is called a round-robin tournament. Such tournaments can be modeled using directed graphs where each team is represented by a vertex. Note that (a, b) is an edge if team a beats team b . This graph is a simple directed graph, containing no loops or multiple directed edges (because no two teams play each other more than once).

Single-Elimination Tournaments. A tournament where each contestant is eliminated after one loss is called a single-elimination tournament. Single-elimination tournaments are often used in sports, including tennis championships and the yearly NCAA basketball championship. We can model such a tournament using a vertex to represent each game and a directed edge to connect a game to the next game the winner of this game played in.

Basic Terminology

- Two vertices u and v in an undirected graph G are called **adjacent** (or **neighbors**) in G if u and v are endpoints of an edge e of G . Such an edge e is called **incident** with the vertices u and v and e is said to **connect** u and v .

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- The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the **neighborhood** of v . If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A . So,

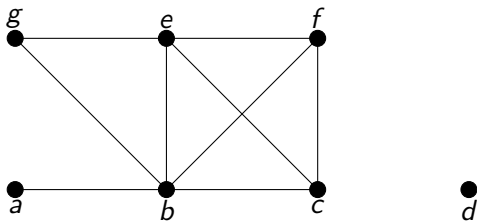
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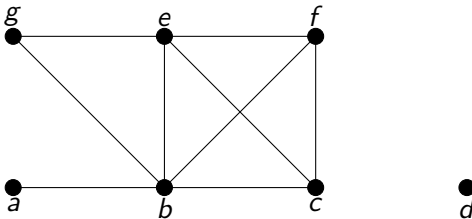
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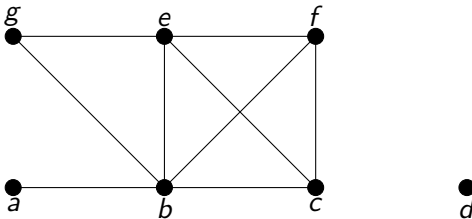
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- By $\deg(v)$ we denote the cardinality of $N(v)$.





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- A vertex is **pendant** if and only if it has degree one. Consequently, a pendant vertex is adjacent to exactly one other vertex. Vertex a in the Example is pendant.

THE HANDSHAKING THEOREM

Theorem

Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v)$$

(Note that this applies even if multiple edges and loops are present.)

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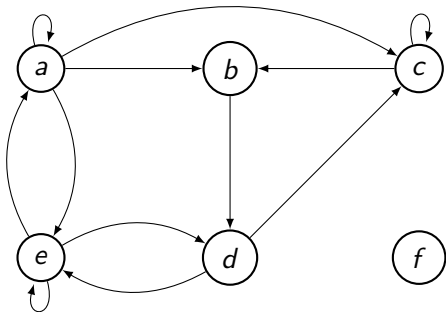
Theorem

An undirected graph has an even number of vertices of odd degree.

- When (u, v) is an edge of the graph G with directed edges, u is said to be **adjacent to** v and v is said to be **adjacent from** u .
- The vertex u is called the **initial vertex** of (u, v) , and v is called the **terminal or end vertex** of (u, v) . The initial vertex and terminal vertex of a loop are the same.
- In a graph with directed edges the **in-degree** of a vertex v , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The **out-degree** of v , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.

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(Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)



Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in the Figure

Theorem

Let $G = (V, E)$ be a graph with directed edges. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

Special Simple Graphs

- **Complete Graphs.** A complete graph on n vertices, denoted by K_n , is a simple graph that contains exactly one edge between each pair of distinct vertices.
- **Cycles.** A cycle C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.
- **Wheels.** We obtain a wheel W_n when we add an additional vertex to a cycle C_n , for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n , by new edges.
- **n -Cubes.** An n -dimensional hypercube, or n -cube, denoted by Q_n , is a graph that has vertices representing the 2^n bit strings of length n . Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position.

Bipartite Graphs

Definition

A simple graph G is called **bipartite** if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a **bipartition** of the vertex set V of G .

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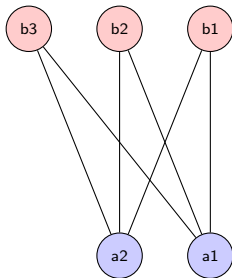
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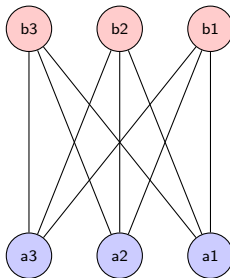
Theorem

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

Complete Bipartite Graphs. A complete bipartite graph $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.



$K_{2,3}$



$K_{3,3}$

Job Assignments. Suppose that there are m employees in a group and n different jobs that need to be done, where $m \geq n$. Each employee is trained to do one or more of these n jobs. We would like to assign an employee to each job. To help with this task, we can use a graph to model employee capabilities. We represent each employee by a vertex and each job by a vertex. For each employee, we include an edge from that employee to all jobs that the employee has been trained to do. Note that the vertex set of this graph can be partitioned into two disjoint sets, the set of employees and the set of jobs, and each edge connects an employee to a job. Consequently, this graph is bipartite, where the bipartition is (E, J) where E is the set of employees and J is the set of jobs.

Finding an assignment of jobs to employees can be thought of as finding a **matching** in the graph model, where a matching M in a simple graph $G = (V, E)$ is a subset of the set E of edges of the graph such that no two edges are incident with the same vertex. In other words, a matching is a subset of edges such that if $\{s, t\}$ and $\{u, v\}$ are distinct edges of the matching, then s , t , u , and v are distinct. A vertex that is the endpoint of an edge of a matching M is said to be **matched in M** ; otherwise it is said to be **unmatched**. A **maximum matching** is a matching with the largest number of edges. We say that a matching M in a bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) is a **complete matching** from V_1 to V_2 if every vertex in V_1 is the endpoint of an edge in the matching, or equivalently, if $|M| = |V_1|$.

HALL'S MARRIAGE THEOREM

Theorem

The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1 .

New Graphs from Old

- A **subgraph** of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$. A subgraph H of G is a **proper subgraph** of G if $H \neq G$.
- Let $G = (V, E)$ be a simple graph. The **subgraph induced** by a subset W of the vertex set V is the graph (W, F) , where the edge set F contains an edge in E if and only if both endpoints of this edge are in W .
- The **union** of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$. The union of G_1 and G_2 is denoted by $G_1 \cup G_2$.